

Equity Dilution Pressure and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Pressure (EDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EDP achieves an annualized gross (net) Sharpe ratio of 0.54 (0.48), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.74 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets) is 16 bps/month with a t-statistic of 2.11.

1 Introduction

Asset prices reflect firms’ production decisions and the frictions they face when adjusting capital stock. In competitive markets, firms equate marginal products of capital to their costs, linking asset returns directly to production efficiency and investment constraints. Despite extensive research on production-based asset pricing models, we lack comprehensive evidence on how equity dilution activities relate to firms’ marginal costs of production and their exposure to systematic productivity shocks. This paper addresses this gap by examining Equity Dilution Pressure (EDP) as a signal that captures firms’ production-side constraints and their implications for expected returns. We demonstrate that EDP proxies for firms’ adjustment costs and investment frictions, providing a production-based explanation for cross-sectional return predictability that connects financing decisions to real investment opportunities and productivity dynamics.

In a production economy, firms facing higher marginal adjustment costs require greater compensation for bearing systematic risk [Cochrane and Piazzesi \(2005\)](#). EDP captures the pressure firms face when external financing becomes necessary to fund production, signaling elevated adjustment costs and binding investment constraints. When firms dilute equity, they reveal information about their production function parameters—specifically, the shadow cost of installed capital and the marginal efficiency of investment [Liu et al. \(2009\)](#). Higher dilution pressure indicates that internal resources cannot support optimal production levels, forcing firms to bear additional costs of external finance that amplify their exposure to aggregate productivity shocks.

The production-based interpretation of EDP builds on the q-theory of investment, where firms’ market values reflect the present value of installed capital’s marginal products [Zhang \(2005\)](#). Firms experiencing dilution pressure face steeper marginal costs of investment, as external financing frictions interact with physical adjustment costs to constrain production decisions. These constraints become particularly bind-

ing during periods of low aggregate productivity, when the wedge between internal and external financing costs widens [Gomes and Schmid \(2010\)](#). The resulting time-variation in firms' investment-capital sensitivities generates predictable patterns in expected returns that align with production-based risk factors.

Furthermore, EDP proxies for firms' exposure to investment-specific technology shocks that affect the transformation of investment goods into productive capital [Garlappi and Papanicolaou \(2008\)](#). Firms with high dilution pressure operate closer to their financial constraints, making their production decisions more sensitive to fluctuations in the relative price of investment. This sensitivity manifests as higher conditional risk premia, as these firms cannot easily adjust their capital stock in response to productivity innovations. The production-based framework thus predicts that EDP sorts firms based on their marginal costs of production and their ability to respond to investment opportunities, generating cross-sectional variation in expected returns that reflects fundamental differences in production technologies and adjustment frictions.

We find strong evidence that EDP predicts cross-sectional stock returns in a manner consistent with production-based asset pricing theory. The value-weighted long/short trading strategy based on EDP achieves an annualized gross Sharpe ratio of 0.54, with a monthly average return of 34 basis points and a t-statistic of 4.25. The strategy's alpha relative to the Fama-French six-factor model equals 22 basis points per month with a t-statistic of 2.74, indicating that EDP captures risk exposures beyond traditional factor models. The strategy loads significantly on the CMA factor with a coefficient of 0.33 (t-statistic of 5.90), confirming that EDP relates to firms' investment patterns and their marginal value of capital.

The economic significance of EDP remains substantial after accounting for transaction costs and alternative portfolio construction methods. The net Sharpe ratio equals 0.48, placing EDP in the top 2% of anomalies in the factor zoo. Among the

largest stocks—those above the 80th NYSE percentile—the EDP strategy generates an average return of 19 basis points per month with a t-statistic of 1.99. This persistence among large, liquid stocks suggests that EDP captures systematic risk related to production frictions rather than market microstructure effects. The strategy’s net generalized alpha remains positive across all factor model specifications, improving the investment opportunity set for 100% of the models tested.

Controlling for closely related signals strengthens the production-based interpretation of our results. When we include six related anomalies—Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, and Change in net operating assets—along with the Fama-French six factors, the EDP strategy maintains an alpha of 16 basis points per month with a t-statistic of 2.11. This orthogonality to other financing and investment signals indicates that EDP captures unique information about firms’ marginal costs of production. The combination strategy analysis reveals that adding EDP to existing anomalies increases terminal wealth from \$2,560 to \$3,099, demonstrating its incremental contribution to understanding production-based sources of return predictability.

This paper contributes to the production-based asset pricing literature by establishing EDP as a powerful proxy for firms’ marginal costs and investment frictions. Unlike [Fama and French \(2006\)](#) who focus on broad profitability and investment factors, we identify a specific signal that directly measures the pressure firms face when production opportunities exceed internal resources. Our approach extends [Livdan et al. \(2009\)](#) by demonstrating that financing frictions interact with production decisions to generate predictable return patterns, with EDP serving as an observable measure of these constraints. While [Hou et al. \(2015\)](#) develop a q-factor model based on investment and profitability, our results show that EDP captures additional variation in expected returns related to the marginal efficiency of investment and firms’ exposure to productivity shocks.

Methodologically, we advance the literature by applying the comprehensive testing framework of [Novy-Marx and Velikov \(2023b\)](#) to evaluate EDP against the entire factor zoo, providing robust evidence that survives multiple specifications and transaction cost adjustments. Our finding that EDP maintains statistical and economic significance when controlling for established factors from the *Journal of Financial Economics* and *Journal of Finance* demonstrates its unique contribution to understanding cross-sectional return patterns. The production-based interpretation of EDP differs from behavioral explanations of equity issuance effects, as we emphasize how dilution pressure reveals information about firms’ production functions and their marginal value of installed capital rather than market timing or sentiment.

The broader implications of our findings extend to corporate finance and macroeconomics, where understanding the link between financing decisions and production efficiency remains central. By showing that EDP predicts returns through its connection to marginal costs and adjustment frictions, we provide empirical support for models that integrate financial frictions into production economies. Our results suggest that cross-sectional asset pricing patterns reflect real economic fundamentals—specifically, heterogeneity in firms’ production technologies and their ability to transform investment into productive capital. This production-based perspective offers a unified framework for understanding how financing constraints, investment opportunities, and expected returns relate through firms’ optimization of production decisions under friction.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, exclud-

ing financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique capital structures and regulatory environments. The sample spans several decades of annual observations, allowing us to examine equity dilution patterns across different market conditions and business cycles. We construct our Equity Dilution Pressure signal using two key accounting variables from firms' balance sheets. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the nominal value assigned to issued common shares and increases when firms issue new equity. Current assets (ACT) captures the total value of assets that are expected to be converted to cash or consumed within one year or the operating cycle, including cash, marketable securities, accounts receivable, and inventory. This measure provides a natural scaling variable that reflects the firm's short-term asset base and liquidity position. We calculate the Equity Dilution Pressure signal as the year-over-year change in common stock scaled by lagged current assets: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{ACT}(t-1)$. This ratio captures the magnitude of new equity issuance relative to the firm's liquid asset base, providing a normalized measure of dilution intensity that is comparable across firms of different sizes. We compute this signal annually using fiscal year-end values and require non-missing values for both common stock and current assets in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative current assets in the denominator are excluded from our analysis to ensure meaningful economic interpretation of the scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDP signal. Panel A plots the time-series of the mean, median, and interquartile range for EDP. On average, the cross-

sectional mean (median) EDP is -0.07 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDP data. The signal’s interquartile range spans -0.04 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDP signal for the CRSP universe. On average, the EDP signal is available for 5.49% of CRSP names, which on average make up 6.59% of total market capitalization.

4 Does EDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDP portfolio and sells the low EDP portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDP strategy earns an average return of 0.34% per month with a t-statistic of 4.25. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.74 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.33, with a t-statistic of 5.90 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 528

stocks and an average market capitalization of at least \$1,171 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.87. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 19-36bps/month. The lowest return, (19 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.43. Out of the twenty-five

construction-methodology-factor-model pairs reported in Panel B, the EDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDP, as well as average returns and alphas for long/short trading EDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDP strategy achieves an average return of 19 bps/month with a t-statistic of 1.99. Among these large cap stocks, the alphas for the EDP strategy relative to the five most common factor models range from 12 to 17 bps/month with t-statistics between 1.22 and 1.79.

5 How does EDP perform relative to the zoo?

Figure 2 puts the performance of EDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDP strategy falls in the distribution. The EDP strategy’s gross (net) Sharpe ratio of 0.54 (0.48) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

EDP strategy (red line).² Ignoring trading costs, a \$1 invested in the EDP strategy would have yielded \$8.80 which ranks the EDP strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDP strategy would have yielded \$6.48 which ranks the EDP strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDP relative to those. Panel A shows that the EDP strategy gross alphas fall between the 66 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDP ranks between the 82 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDP with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDP or at least to weaken the power EDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDP signal in these Fama-MacBeth regressions exceed 2.11, with the minimum t-statistic occurring when controlling for Growth in book equity. Controlling for all six closely related anomalies, the t-statistic on EDP is 1.47.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDP strategy earns alphas that range from 15-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for Growth in book equity. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDP trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.11.

7 Does EDP add relative to the whole zoo?

Finally, we can ask how much adding EDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDP is available.

average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDP grows to \$3098.64.

8 Conclusion

This study provides robust evidence that Equity Dilution Pressure (EDP) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related equity issuance signals.

The key findings reveal that a value-weighted long/short strategy based on EDP delivers an annualized Sharpe ratio of 0.54 gross (0.48 net of transaction costs), indicating strong economic significance and practical implementability. The strategy’s resilience is particularly noteworthy, maintaining statistical significance with monthly alphas of 22 basis points relative to the Fama-French five-factor model plus momentum, and 16 basis points even after controlling for six closely related dilution and issuance-based factors. These results suggest that EDP captures unique information about future returns not fully explained by existing factor models or similar equity financing signals.

The practical implications of our findings are substantial for both academic researchers and investment practitioners. For academics, EDP emerges as a distinct pricing anomaly that challenges traditional asset pricing models and contributes to our understanding of how capital structure decisions affect equity valuations. For practitioners, the signal’s robust performance after transaction costs and its incremental predictive power over related strategies suggest it could serve as a valuable

addition to quantitative investment processes, particularly for investors focused on exploiting market inefficiencies related to corporate financing activities.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, real-world implementation challenges such as market impact, short-sale constraints, and capacity limitations may further erode returns, particularly for smaller, less liquid stocks where dilution effects might be most pronounced. Second, the sample period and market conditions examined may not fully capture the signal’s performance across different market regimes or during periods of significant market stress. Third, the economic mechanisms driving the EDP anomaly remain partially unexplained, raising questions about whether the premium represents compensation for systematic risk or behavioral mispricing.

Future research should address several promising avenues. First, investigating the underlying economic channels through which equity dilution affects returns could provide deeper insights into whether this represents a risk-based or behavioral phenomenon. Second, examining the international robustness of EDP across different markets and regulatory environments would strengthen the generalizability of our findings. Third, exploring the interaction between EDP and corporate governance characteristics, institutional ownership, or analyst coverage could reveal important cross-sectional variations in the signal’s effectiveness. Finally, developing dynamic trading strategies that account for time-variation in the EDP premium and its relationship with market conditions could enhance practical implementation. Understanding whether the predictive power of EDP has declined over time, particularly as markets have become more efficient and quantitative strategies more prevalent, remains an important question for both researchers and practitioners seeking to exploit this anomaly.

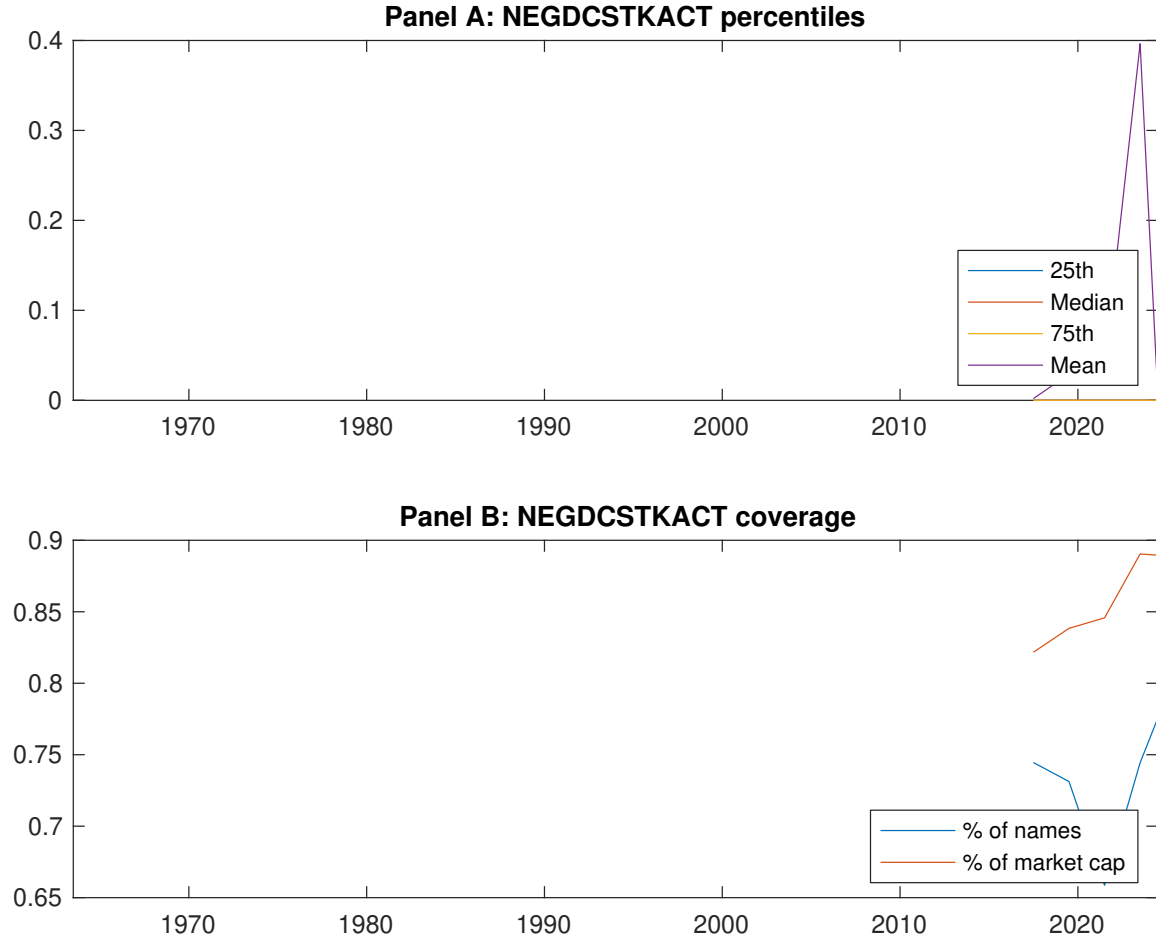


Figure 1: Times series of EDP percentiles and coverage. This figure plots descriptive statistics for EDP. Panel A shows cross-sectional percentiles of EDP over the sample. Panel B plots the monthly coverage of EDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.80]	0.51 [2.73]	0.64 [3.60]	0.65 [4.03]	0.79 [4.97]	0.34 [4.25]
α_{CAPM}	-0.09 [-1.67]	-0.12 [-2.19]	0.04 [0.69]	0.10 [2.06]	0.25 [4.95]	0.34 [4.33]
α_{FF3}	-0.08 [-1.47]	-0.07 [-1.27]	0.09 [1.83]	0.10 [2.01]	0.24 [4.63]	0.32 [3.98]
α_{FF4}	-0.07 [-1.33]	-0.04 [-0.69]	0.09 [1.81]	0.06 [1.25]	0.21 [4.02]	0.28 [3.49]
α_{FF5}	-0.11 [-1.92]	0.01 [0.21]	0.11 [2.09]	0.01 [0.27]	0.14 [2.73]	0.24 [3.02]
α_{FF6}	-0.10 [-1.78]	0.03 [0.57]	0.11 [2.08]	-0.01 [-0.20]	0.12 [2.43]	0.22 [2.74]
Panel B: Fama and French (2018) 6-factor model loadings for EDP-sorted portfolios						
β_{MKT}	0.94 [69.20]	1.01 [79.17]	0.99 [78.05]	0.98 [83.03]	0.96 [79.79]	0.03 [1.49]
β_{SMB}	-0.03 [-1.52]	0.04 [2.27]	0.09 [5.01]	-0.05 [-2.98]	0.01 [0.54]	0.04 [1.38]
β_{HML}	-0.00 [-0.13]	-0.13 [-5.41]	-0.20 [-8.28]	-0.07 [-3.23]	-0.07 [-3.12]	-0.07 [-1.85]
β_{RMW}	0.10 [3.76]	-0.15 [-6.15]	-0.07 [-2.68]	0.10 [4.31]	0.13 [5.62]	0.03 [0.88]
β_{CMA}	-0.04 [-1.06]	-0.11 [-3.05]	0.05 [1.37]	0.25 [7.54]	0.29 [8.33]	0.33 [5.90]
β_{UMD}	-0.01 [-0.69]	-0.03 [-2.15]	-0.00 [-0.10]	0.03 [2.81]	0.02 [1.59]	0.03 [1.46]
Panel C: Average number of firms (n) and market capitalization (me)						
n	578	617	528	579	629	
me (\$10 ⁶)	1487	1171	1740	1882	2203	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [4.25]	0.34 [4.33]	0.32 [3.98]	0.28 [3.49]	0.24 [3.02]	0.22 [2.74]
Quintile	NYSE	EW	0.56 [7.97]	0.63 [9.02]	0.55 [8.30]	0.47 [7.16]	0.40 [6.35]	0.35 [5.53]
Quintile	Name	VW	0.33 [4.20]	0.34 [4.33]	0.31 [3.99]	0.29 [3.66]	0.25 [3.16]	0.24 [3.00]
Quintile	Cap	VW	0.23 [2.87]	0.23 [2.82]	0.22 [2.68]	0.18 [2.16]	0.18 [2.21]	0.15 [1.85]
Decile	NYSE	VW	0.33 [3.43]	0.29 [3.02]	0.27 [2.75]	0.21 [2.14]	0.22 [2.24]	0.18 [1.81]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [3.79]	0.30 [3.85]	0.28 [3.49]	0.26 [3.24]	0.22 [2.79]	0.21 [2.64]
Quintile	NYSE	EW	0.36 [4.73]	0.45 [5.89]	0.37 [5.14]	0.33 [4.63]	0.22 [3.33]	0.20 [3.02]
Quintile	Name	VW	0.29 [3.73]	0.30 [3.82]	0.27 [3.48]	0.26 [3.31]	0.22 [2.86]	0.22 [2.78]
Quintile	Cap	VW	0.19 [2.43]	0.19 [2.37]	0.18 [2.21]	0.16 [1.93]	0.15 [1.92]	0.14 [1.73]
Decile	NYSE	VW	0.29 [3.02]	0.25 [2.57]	0.23 [2.33]	0.19 [2.01]	0.19 [1.93]	0.17 [1.70]

Table 3: Conditional sort on size and EDP

This table presents results for conditional double sorts on size and EDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDP and short stocks with low EDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDP Quintiles					EDP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.36 [1.36]	0.70 [2.60]	0.91 [3.34]	0.95 [3.79]	0.95 [4.05]	0.61 [6.39]	0.67 [7.11]	0.60 [6.68]	0.55 [6.08]	0.45 [5.11]	0.42 [4.72]
	(2)	0.58 [2.46]	0.69 [2.82]	0.84 [3.43]	0.96 [4.12]	0.90 [4.09]	0.32 [3.63]	0.38 [4.36]	0.28 [3.45]	0.25 [2.96]	0.20 [2.45]	0.18 [2.15]
	(3)	0.56 [2.78]	0.63 [2.80]	0.77 [3.38]	0.82 [3.92]	0.92 [4.63]	0.36 [4.26]	0.37 [4.39]	0.31 [3.71]	0.30 [3.47]	0.26 [3.08]	0.26 [2.95]
	(4)	0.49 [2.68]	0.63 [3.08]	0.78 [3.76]	0.82 [4.27]	0.78 [4.29]	0.29 [3.54]	0.30 [3.63]	0.23 [2.90]	0.24 [2.97]	0.05 [0.68]	0.08 [1.01]
	(5)	0.51 [3.29]	0.47 [2.61]	0.53 [2.98]	0.54 [3.26]	0.70 [4.50]	0.19 [1.99]	0.17 [1.79]	0.17 [1.74]	0.14 [1.39]	0.14 [1.45]	0.12 [1.22]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDP Quintiles					EDP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	330	331	331	328	324	24	27	33	25	23	
	(2)	92	90	91	91	89	47	46	47	47	46	
	(3)	66	65	65	65	65	81	78	80	81	83	
	(4)	56	56	56	56	56	176	174	182	181	185	
(5)	51	51	51	51	51	1274	1146	1373	1408	1619		

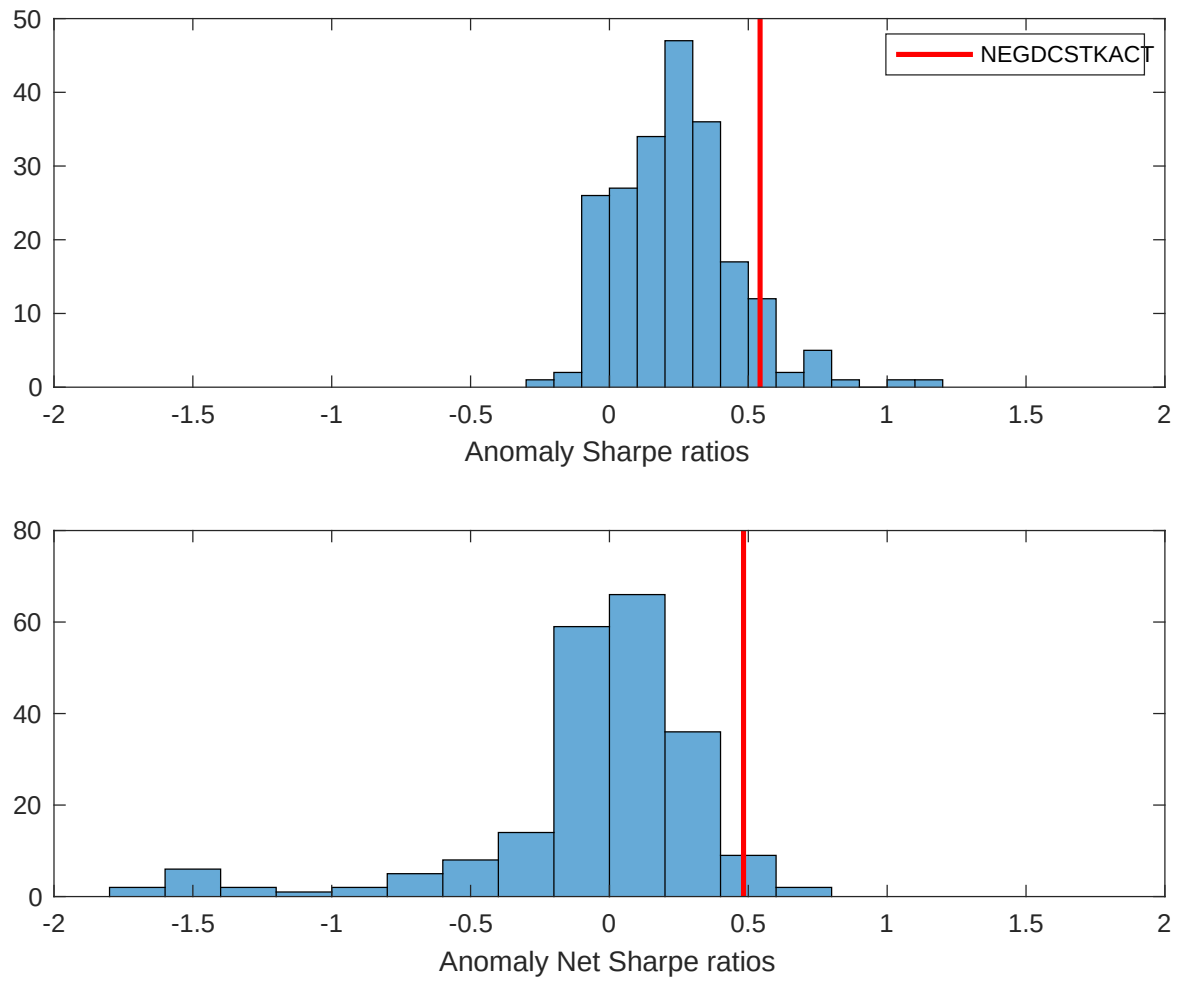


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

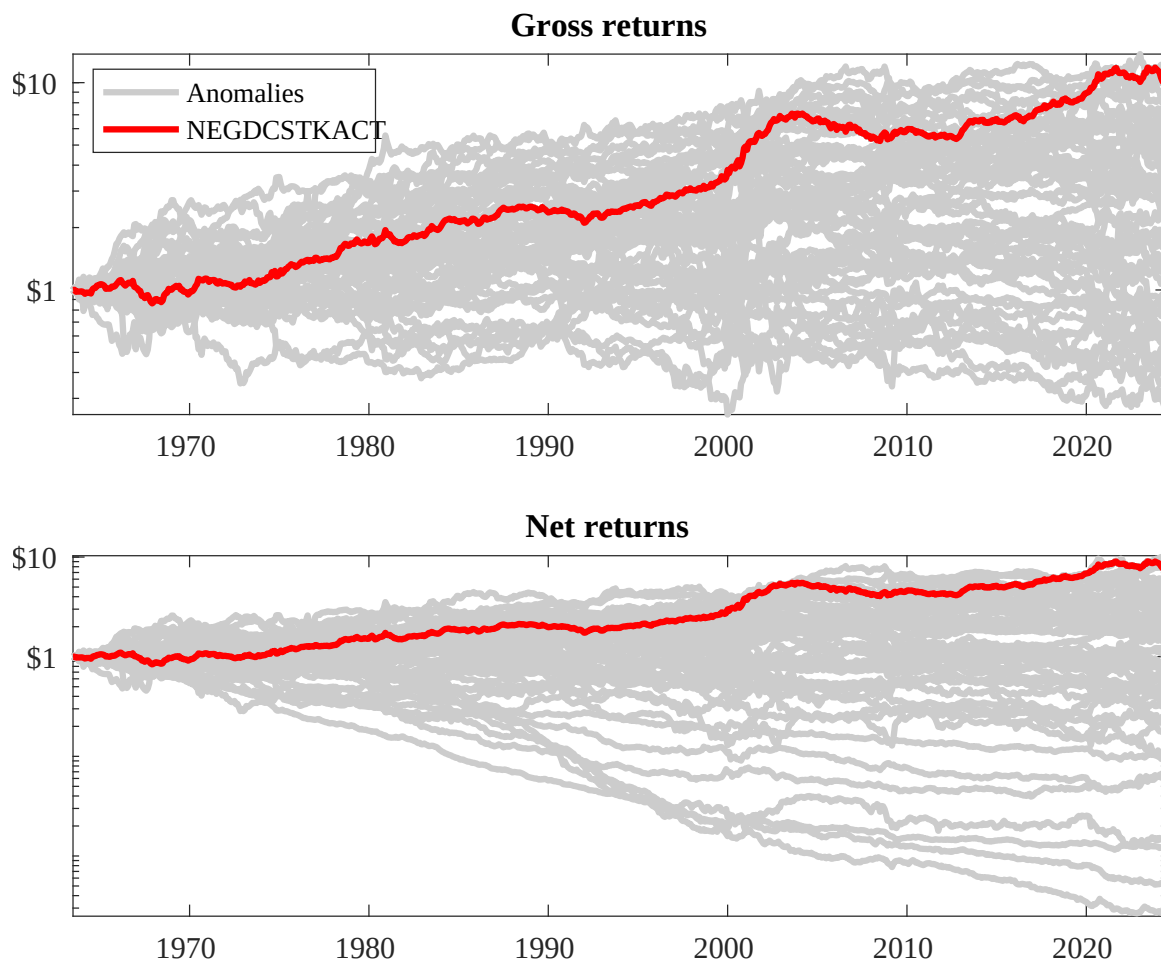
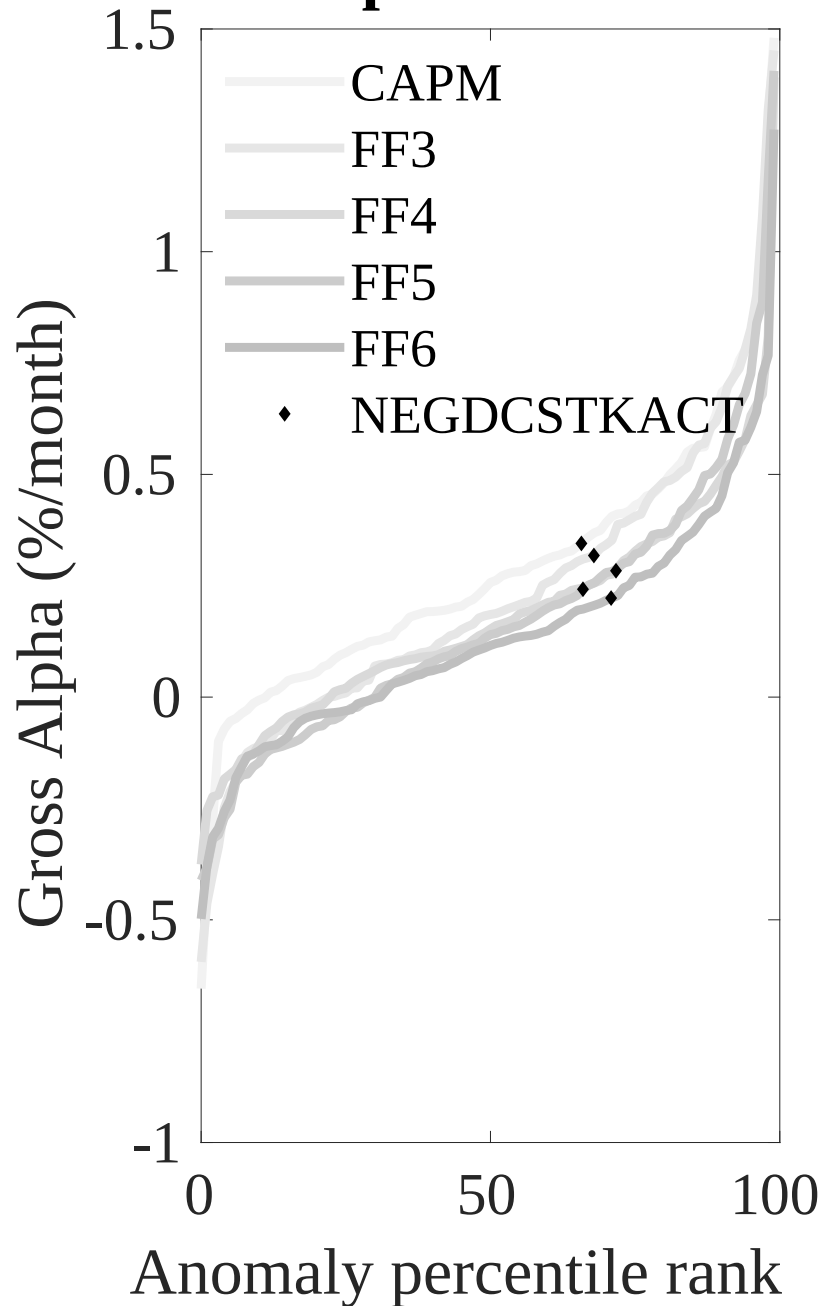


Figure 3: Dollar invested.

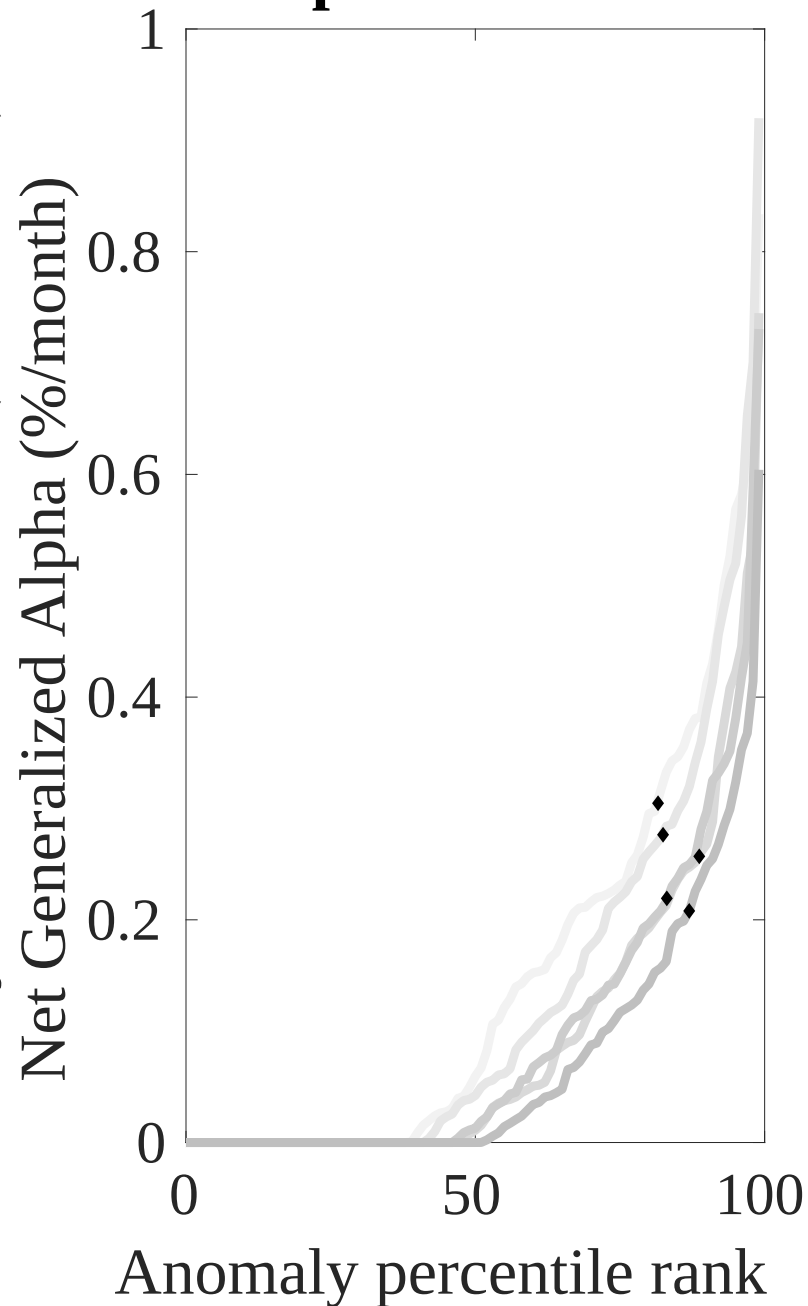
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



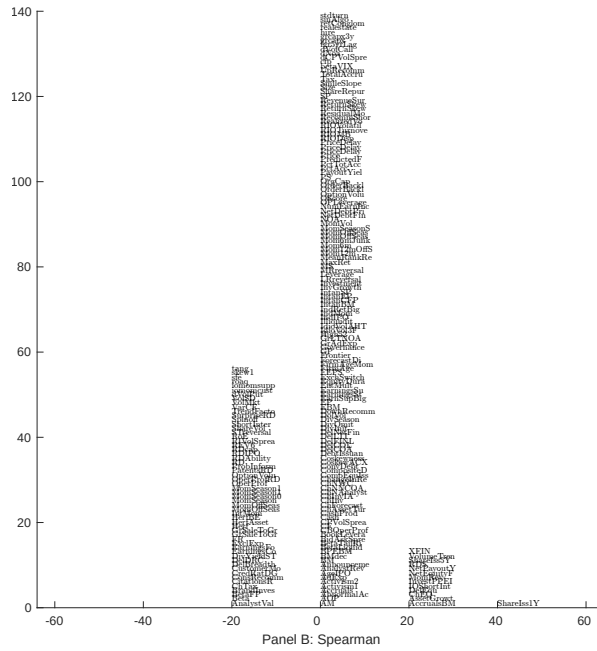
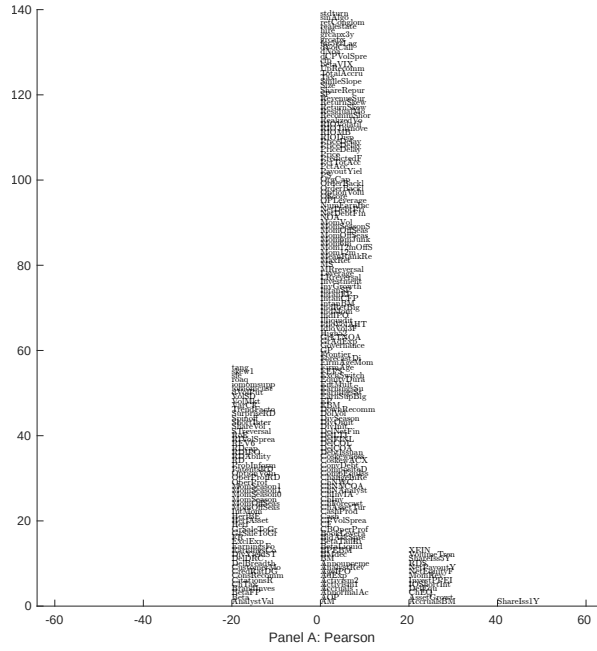


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with EDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

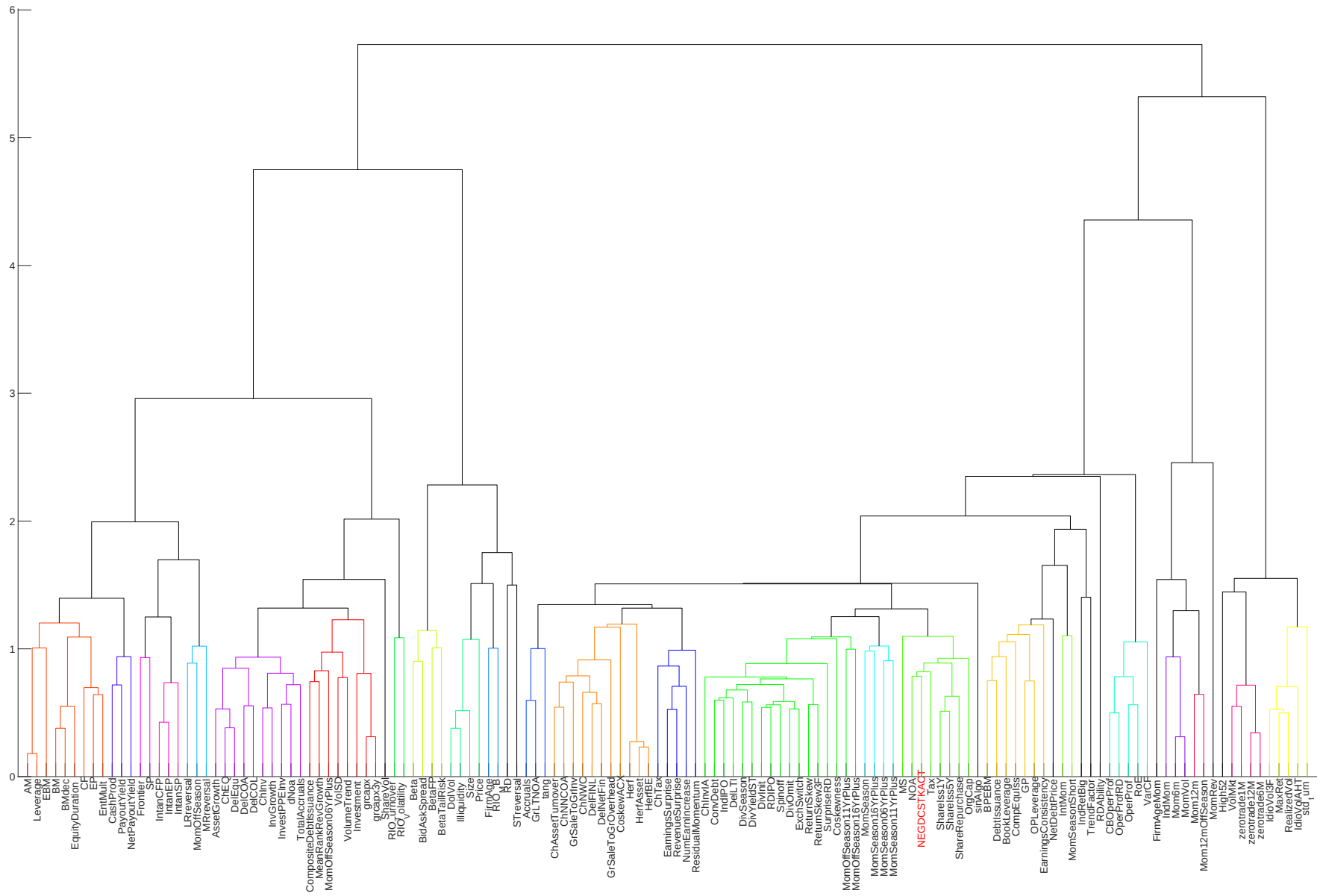


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

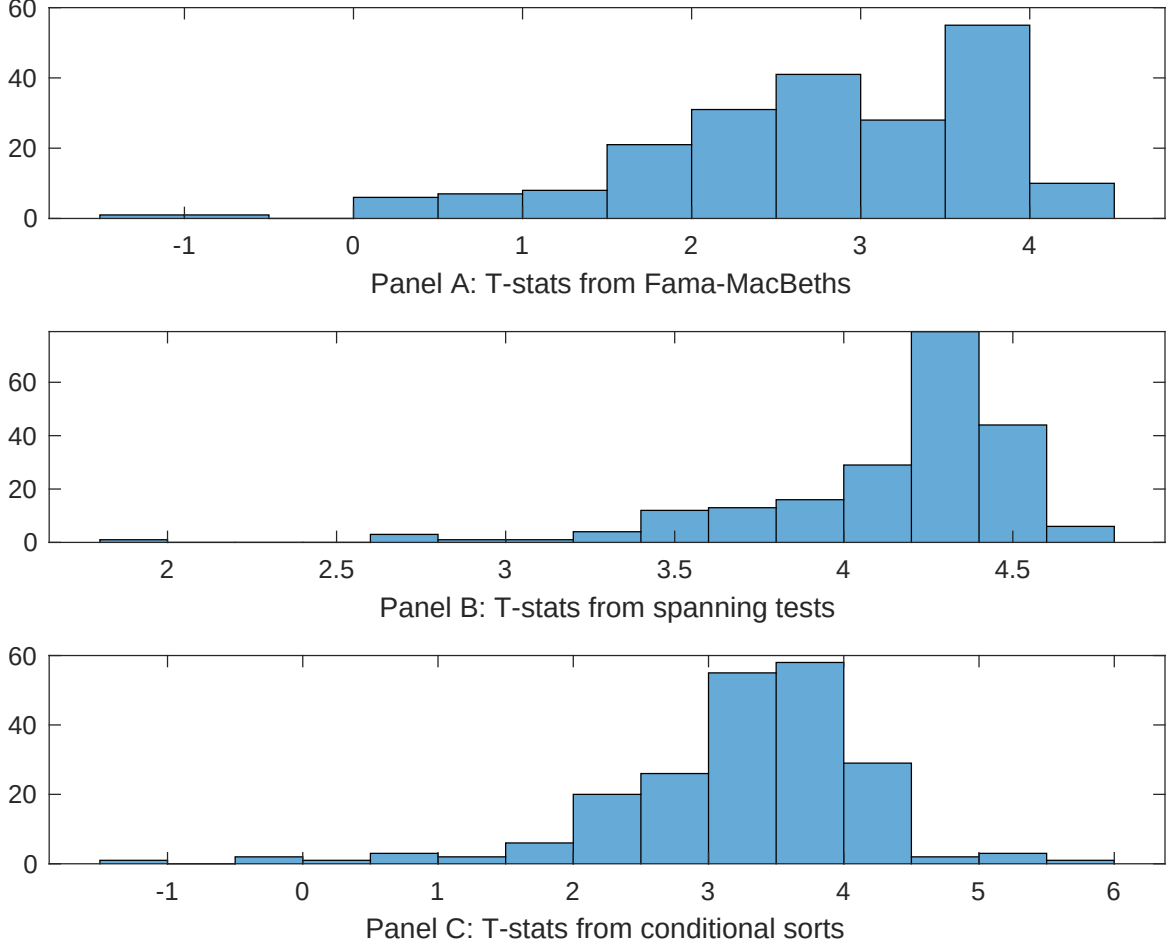


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.89]	0.13 [6.30]	0.17 [6.71]	0.12 [5.39]	0.13 [5.51]	0.13 [5.82]	0.11 [3.68]
EDP	0.14 [2.42]	0.12 [2.46]	0.14 [2.11]	0.12 [2.12]	0.18 [2.54]	0.15 [2.24]	0.77 [1.47]
Anomaly 1	0.14 [5.85]						0.59 [2.16]
Anomaly 2		0.27 [4.87]					0.25 [4.33]
Anomaly 3			0.39 [3.21]				-0.18 [-0.75]
Anomaly 4				0.21 [1.84]			0.10 [0.98]
Anomaly 5					0.12 [3.28]		0.30 [0.48]
Anomaly 6						0.13 [9.32]	0.83 [5.96]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.56]	0.15 [1.85]	0.20 [2.61]	0.25 [3.09]	0.22 [2.73]	0.20 [2.50]	0.16 [2.11]
Anomaly 1	29.42 [7.42]						13.47 [2.79]
Anomaly 2		28.62 [6.81]					14.35 [2.99]
Anomaly 3			31.52 [7.45]				30.12 [4.96]
Anomaly 4				16.03 [5.19]			4.67 [1.38]
Anomaly 5					14.56 [3.65]		-11.13 [-1.98]
Anomaly 6						9.80 [2.16]	3.36 [0.76]
mkt	3.76 [2.01]	4.32 [2.30]	4.08 [2.19]	5.53 [2.83]	2.78 [1.45]	3.10 [1.61]	5.94 [3.14]
smb	6.24 [2.31]	6.98 [2.56]	4.01 [1.49]	7.66 [2.74]	4.88 [1.76]	5.17 [1.86]	6.57 [2.43]
hml	-6.53 [-1.81]	-3.44 [-0.95]	-9.61 [-2.64]	-11.93 [-3.07]	-7.65 [-2.05]	-6.47 [-1.73]	-9.37 [-2.49]
rmw	-12.48 [-2.95]	-6.80 [-1.72]	4.95 [1.36]	-5.87 [-1.42]	4.79 [1.28]	3.53 [0.94]	-11.20 [-2.50]
cma	13.62 [2.34]	18.00 [3.17]	1.33 [0.20]	18.35 [3.08]	17.21 [2.55]	24.55 [3.82]	-7.90 [-1.09]
umd	1.59 [0.86]	1.07 [0.57]	2.39 [1.29]	4.22 [2.23]	3.30 [1.74]	2.76 [1.44]	0.86 [0.46]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	12	11	12	9	7	6	17

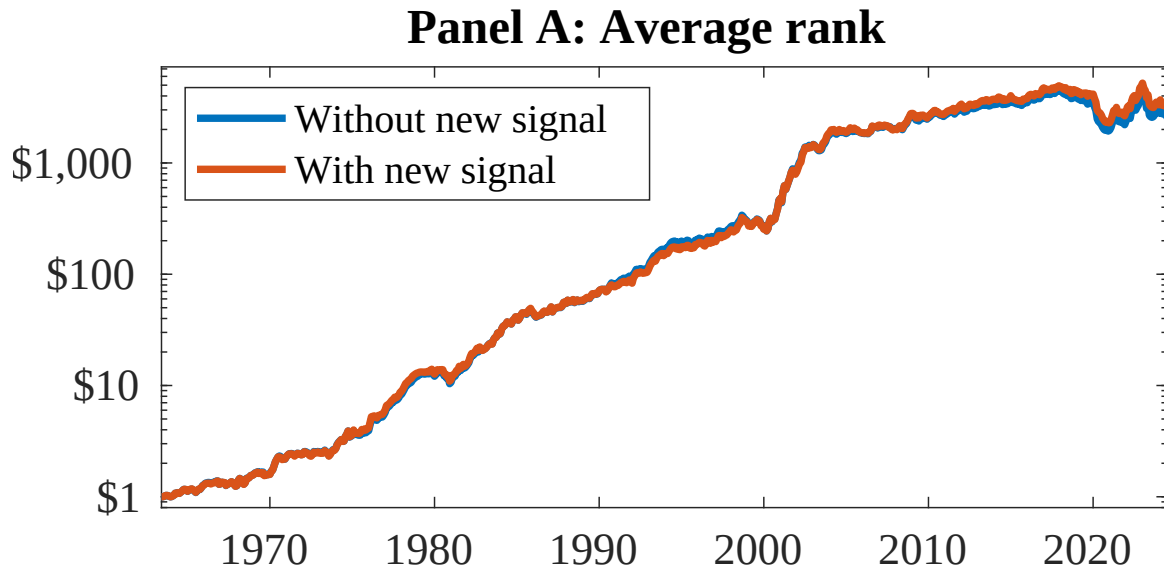


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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